

DFIR Investigation Report

Operation « Stonks »

Windows Data Deduplication forensics and recovery
of a deleted financial report

Case:	Operation « Stonks », host FSERVER01
Evidence:	evidence.ad1 (AccessData AD1 v4, logical image)
Date:	05/07/2026
Classification:	Confidential

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1. Executive Summary

This report documents the forensic investigation of **evidence.ad1**, a logical disk image acquired from **FSERVER01**, a Windows Server configured in the SE Asia Standard Time (UTC+7) time zone. The objective was to recover a deleted financial report and to establish whether the company's 2024 accounts had been falsified.

The investigation established that the organisation maintained two parallel versions of its 2024 Consolidated Financial Statements: a publicly presented, audited version reporting a profit, and an internal version, authored by a member of the accounting team, revealing a real loss. The internal (truthful) document was deleted to conceal the losses.

Because Windows Data Deduplication was active on the D: volume, the deleted file's content still resided in the Deduplication ChunkStore. The file was fully reconstructed from its constituent chunks and validated by cryptographic hash, allowing a line-by-line comparison of both income statements.

Item	Finding
Compromised host	FSERVER01 (Windows Server, UTC+7)
Evidence container	evidence.ad1, AccessData AD1 v4 (logical, 166 MB)
Volumes of interest	C: (system, logs, tasks, registry) and D: 'Data' (Finance, HR, Dedup store)
Key artefact	Deleted internal 2024 financial statements (.docx) on D:\Finance\Reports\Annual\2024
Recovery method	Windows Dedup ChunkStore reconstruction driven by the NTFS reparse point
Nature of the case	Falsification of financial statements and deletion of material evidence

2. Methodology & Evidence Handling

2.1 Investigative approach

The investigation followed a structured, repeatable DFIR methodology: verification of evidence integrity, reconnaissance of the acquired volumes, timeline reconstruction from Windows event logs and Deduplication artefacts, analysis of the NTFS Master File Table (\$MFT) to identify the deleted file, and finally reconstruction of the file content from the Deduplication ChunkStore. Every analytical step is supported by a screenshot captured directly in the analysis tooling.

2.2 Tooling

- **FTK Imager 8.2.0.59** (Exterro): mounting the AD1 image, browsing NTFS structures, hex inspection and file export.
- **Windows Event Viewer**: examination of the Setup and Deduplication operational logs.
- **PowerShell**: archive extraction and metadata inspection.
- **recover.py** (custom Python tool developed for this case): automated reconstruction of a deduplicated file from the ChunkStore, driven by the NTFS reparse point.

2.3 Time-zone handling

Windows event logs (.evtx) and NTFS FILETIME values are stored in **UTC**. The host time zone is **SE Asia Standard Time (UTC+7)**, read from the SYSTEM registry hive (ControlSet001\Control\TimeZoneInformation\TimeZoneKeyName = "SE Asia Standard Time", Bias = -420 min). Timestamps are therefore always interpreted in both UTC and local time to avoid ambiguity.

2.4 Evidence integrity (chain of custody)

The image was verified in FTK Imager. The computed hashes below establish the integrity baseline of the evidence container for the remainder of the investigation.

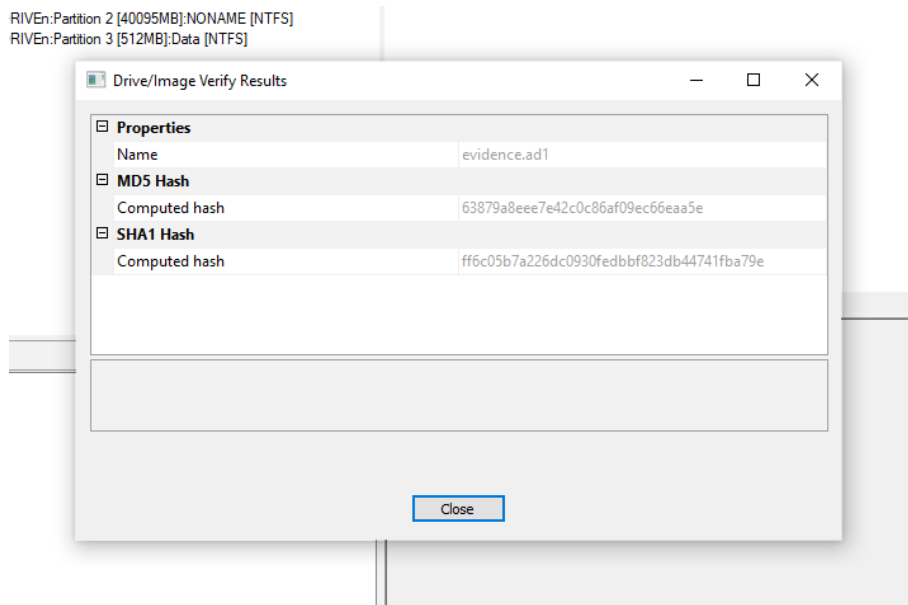


Figure 2.1: FTK Imager "Drive/Image Verify Results" for evidence.ad1.

Algorithm	Computed hash
MD5	63879a8eee7e42c0c86af09ec66eaa5e
SHA-1	ff6c05b7a226dc0930fedbbf823db44741fba79e

3. Reconnaissance of the Evidence

The container was identified as an **AccessData AD1 v4** logical image (magic ADSEGMENTEDFILE / ADLOGICALIMAGE). When opened in FTK Imager, the evidence tree exposes two NTFS partitions.

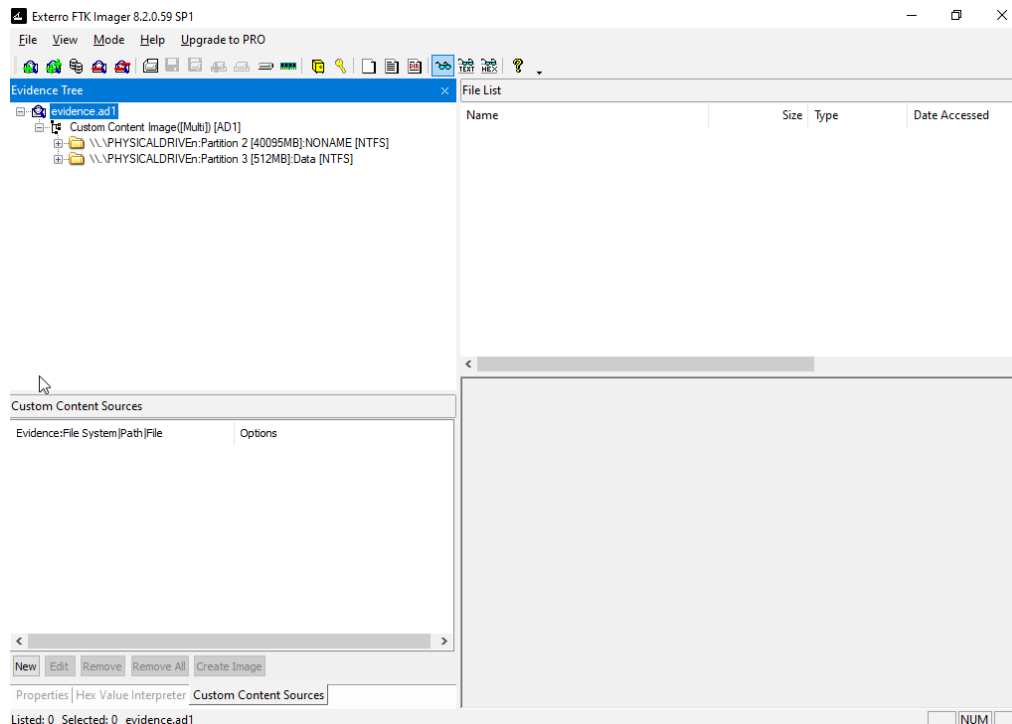


Figure 3.1: Evidence tree: Partition 2 (C:) and Partition 3 (D: 'Data').

Observation. The image contains **two sibling NTFS volumes**. Partition 2 is the system volume **C:** (event logs, scheduled tasks, registry hives). Partition 3, labelled '**Data**' (**D:**), holds the material of the investigation: the Finance/ and HR/ directories and, above all, System Volume Information\\Dedup\\ (the Deduplication ChunkStore, Settings, State and Logs). Both volumes must be examined; the deleted financial report lives on D:.

4. Investigation: Detailed Findings

The following sections address each investigative question in turn. Each is documented with its source artefact, the analytical reasoning and supporting screenshots.

4.1 Installation of Data Deduplication

Source: C:\Windows\System32\winevt\Logs\Setup.evtx, examined in Event Viewer. The Servicing provider records the installation of the Deduplication package: **Event ID 7** marks the start of the package installation and **Event ID 9** marks its successful completion (ErrorCode 0x0).

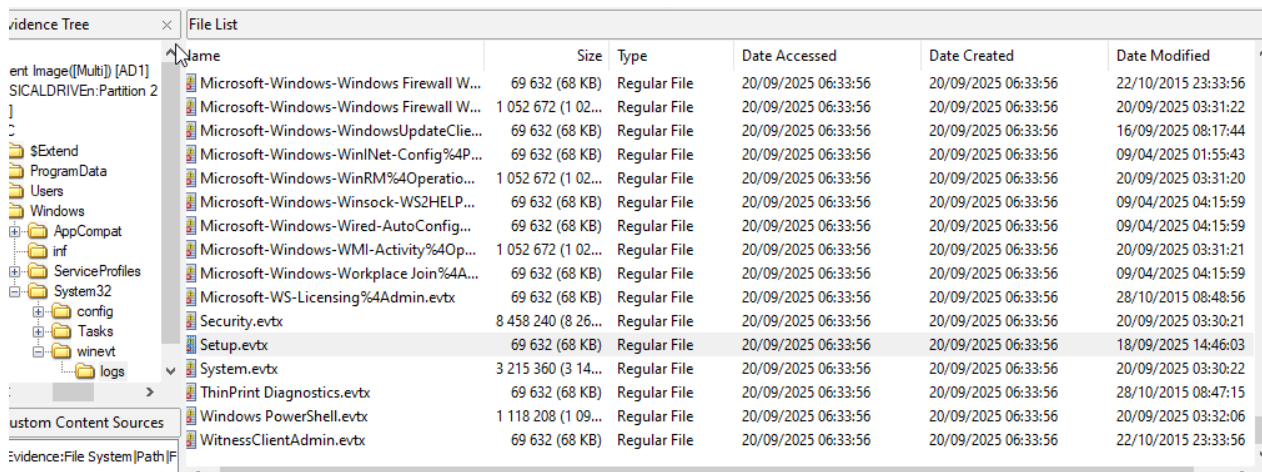


Figure 4.1: Setup.evtx selected in FTK Imager (C:\Windows\System32\winevt\Logs).

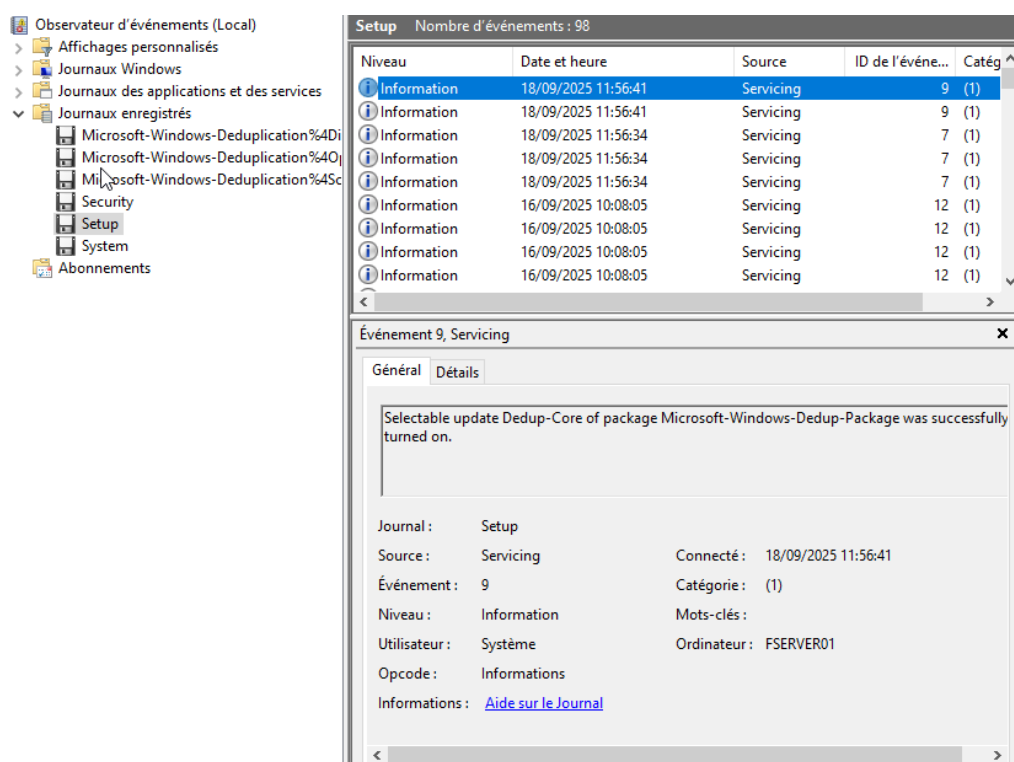
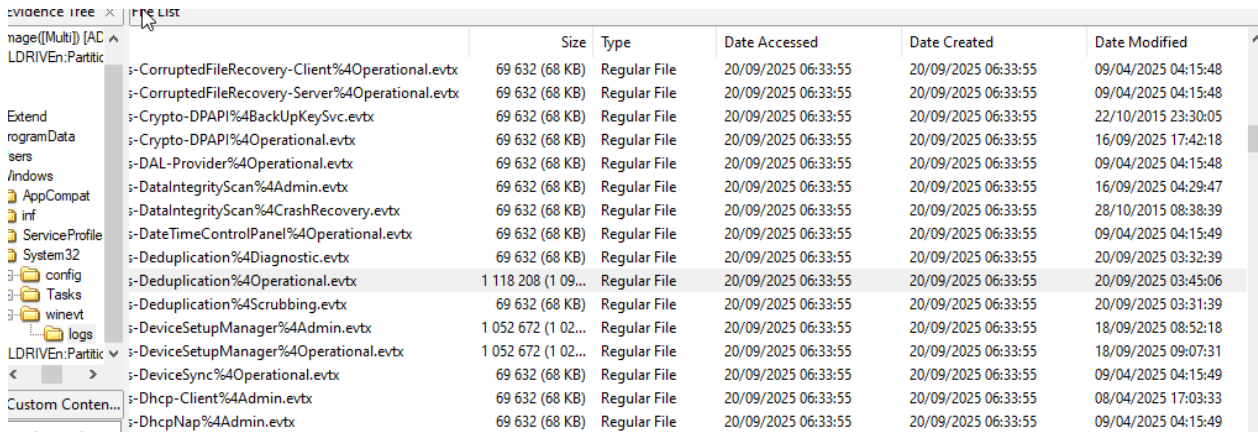


Figure 4.2: Event 9 (Servicing): the Dedup-Core package was successfully turned on, host FSERVER01 (time displayed in the analyst's local time zone).

The first Event ID 9 in chronological order gives the first installation of the feature. As Event Viewer renders timestamps in the analyst workstation's time zone, the value must be normalised to UTC before being converted to the host's local time (UTC+7).

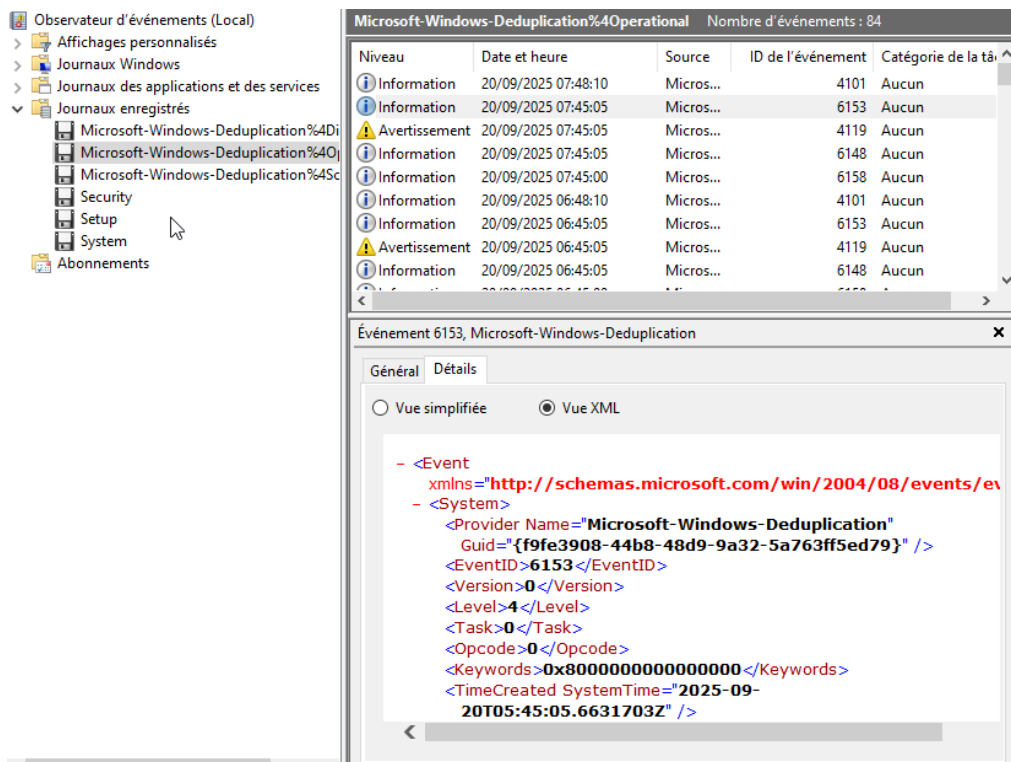
4.2 Disk space saved by the first optimisation job

Source: Microsoft-Windows-Deduplication%4Operational.evtx, **Event ID** 6153 (optimisation job completed). The first chronological 6153 event reports the saved space for volume D: in its XML payload (SavedSpace, SavingsRate, OptimizedFileCount).



File Name	Size	Type	Date Accessed	Date Created	Date Modified
-CorruptedFileRecovery-Client%4Operational.evtx	69 632 (68 KB)	Regular File	20/09/2025 06:33:55	20/09/2025 06:33:55	09/04/2025 04:15:48
-CorruptedFileRecovery-Server%4Operational.evtx	69 632 (68 KB)	Regular File	20/09/2025 06:33:55	20/09/2025 06:33:55	09/04/2025 04:15:48
-Crypto-DPAPI%4BackupKeySvc.evtx	69 632 (68 KB)	Regular File	20/09/2025 06:33:55	20/09/2025 06:33:55	22/10/2015 23:30:05
-Crypto-DPAPI%4Operational.evtx	69 632 (68 KB)	Regular File	20/09/2025 06:33:55	20/09/2025 06:33:55	16/09/2025 17:42:18
-DAL-Provider%4Operational.evtx	69 632 (68 KB)	Regular File	20/09/2025 06:33:55	20/09/2025 06:33:55	09/04/2025 04:15:48
-DataIntegrityScan%4Admin.evtx	69 632 (68 KB)	Regular File	20/09/2025 06:33:55	20/09/2025 06:33:55	16/09/2025 04:29:47
-DataIntegrityScan%4CrashRecovery.evtx	69 632 (68 KB)	Regular File	20/09/2025 06:33:55	20/09/2025 06:33:55	28/10/2015 08:38:39
-DateTimeControlPanel%4Operational.evtx	69 632 (68 KB)	Regular File	20/09/2025 06:33:55	20/09/2025 06:33:55	09/04/2025 04:15:49
-Deduplication%4Diagnostic.evtx	69 632 (68 KB)	Regular File	20/09/2025 06:33:55	20/09/2025 06:33:55	20/09/2025 03:32:39
-Deduplication%4Operational.evtx	1 118 208 (1 09...)	Regular File	20/09/2025 06:33:55	20/09/2025 06:33:55	20/09/2025 03:45:06
-Deduplication%4Scrubbing.evtx	69 632 (68 KB)	Regular File	20/09/2025 06:33:55	20/09/2025 06:33:55	20/09/2025 03:31:39
-DeviceSetupManager%4Admin.evtx	1 052 672 (1 02...)	Regular File	20/09/2025 06:33:55	20/09/2025 06:33:55	18/09/2025 08:52:18
-DeviceSetupManager%4Operational.evtx	1 052 672 (1 02...)	Regular File	20/09/2025 06:33:55	20/09/2025 06:33:55	18/09/2025 09:07:31
-DeviceSync%4Operational.evtx	69 632 (68 KB)	Regular File	20/09/2025 06:33:55	20/09/2025 06:33:55	09/04/2025 04:15:49
-Dhcp-Client%4Admin.evtx	69 632 (68 KB)	Regular File	20/09/2025 06:33:55	20/09/2025 06:33:55	08/04/2025 17:03:33
-DhcpNap%4Admin.evtx	69 632 (68 KB)	Regular File	20/09/2025 06:33:55	20/09/2025 06:33:55	09/04/2025 04:15:49

Figure 4.3: Deduplication operational log selected in the winevt\Logs directory.



Observateur d'événements (Local)

- Affichages personnalisés
- Journaux Windows
- Journaux des applications et des services
- Journaux enregistrés
 - Microsoft-Windows-Deduplication%4Di...
 - Microsoft-Windows-Deduplication%4O...
 - Microsoft-Windows-Deduplication%4Sc...
 - Security
 - Setup
 - System
 - Abonnements

Microsoft-Windows-Deduplication%4Operational Nombre d'événements : 84

Niveau	Date et heure	Source	ID de l'événement	Catégorie de la tâche
Information	20/09/2025 07:48:10	Micros...	4101	Aucun
Information	20/09/2025 07:45:05	Micros...	6153	Aucun
Avertissement	20/09/2025 07:45:05	Micros...	4119	Aucun
Information	20/09/2025 07:45:05	Micros...	6148	Aucun
Information	20/09/2025 07:45:00	Micros...	6158	Aucun
Information	20/09/2025 06:48:10	Micros...	4101	Aucun
Information	20/09/2025 06:45:05	Micros...	6153	Aucun
Avertissement	20/09/2025 06:45:05	Micros...	4119	Aucun
Information	20/09/2025 06:45:05	Micros...	6148	Aucun

Événement 6153, Microsoft-Windows-Deduplication

Général Détails

☐ Vue simplifiée ☒ Vue XML

```

- <Event
  xmlns="http://schemas.microsoft.com/win/2004/08/events/event"
- <System>
  <Provider Name="Microsoft-Windows-Deduplication"
    Guid="{f9fe3908-44b8-48d9-9a32-5a763ff5ed79}" />
  <EventID>6153</EventID>
  <Version>0</Version>
  <Level>4</Level>
  <Task>0</Task>
  <Opcode>0</Opcode>
  <Keywords>0x8000000000000000</Keywords>
  <TimeCreated SystemTime="2025-09-20T05:45:05.6631703Z" />
  
```

Figure 4.4: Event 6153 (Microsoft-Windows-Deduplication), XML detail view.

4.3 Last re-activation of Deduplication on D:

Source: D:\System Volume Information\Dedup\Settings\dedupConfig.02.xml. The changeTime field (a FILETIME, stored UTF-16) records the last modification of the volume configuration, i.e. the last time deduplication settings were (re)applied. The decoded value is corroborated in the operational log by a manual optimisation job (JobPriorityType 3) fourteen seconds later, following the classic Enable-DedupVolume D: then Start-DedupJob sequence.

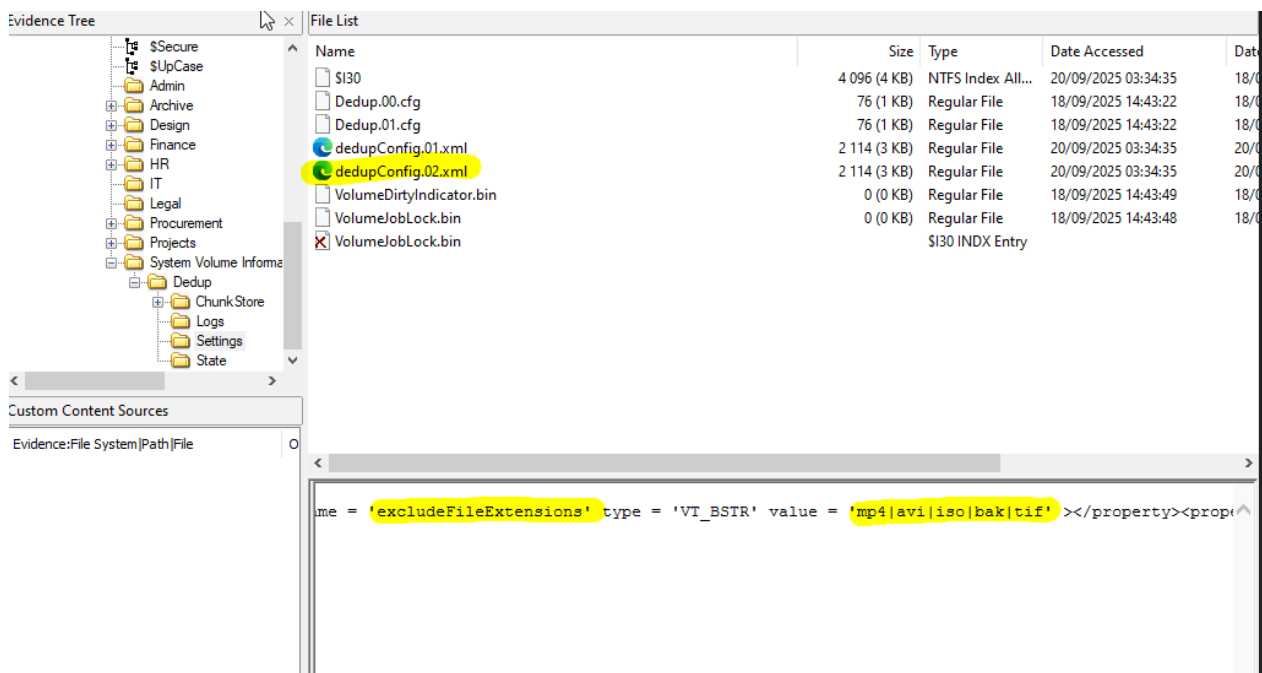


Figure 4.5: dedupConfig.02.xml in the Dedup\Settings directory; the excludeFileExtensions property is highlighted.

4.4 File extensions excluded from optimisation

Source: dedupConfig.02.xml. Two properties list excluded extensions: excludeFileExtensions (administrator-defined, visible in Figure 4.5) and excludeFileExtensionsDefault (system defaults). The complete exclusion set is the union of both lists.

Note. The noCompressionFileExtensions list (zip, rar, ...) is deliberately not treated as an exclusion: those files are optimised by deduplication, they are merely stored without additional compression. This distinction later explains why the recovered .docx chunks are stored uncompressed.

4.5 Number of optimised files

Source: D:\System Volume Information\Dedup\State\dedupStatistics.xml, property optimizedFileCount. The value is confirmed by the last 6153 optimisation event.

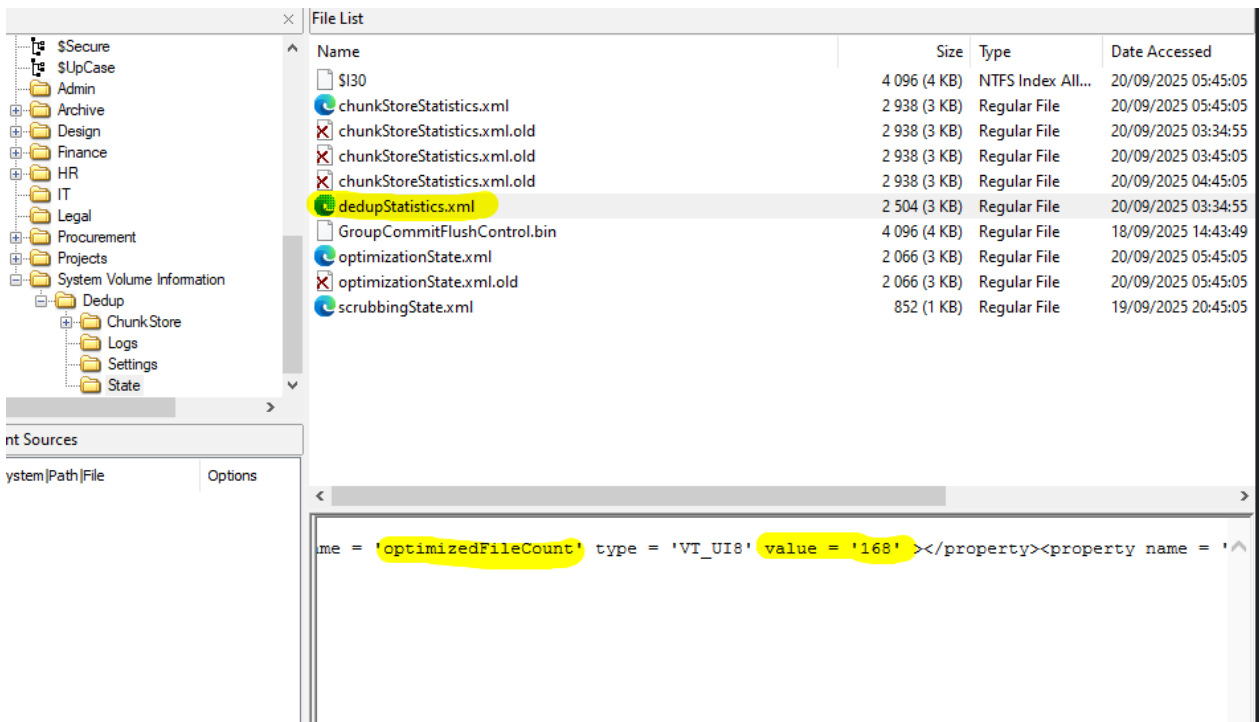


Figure 4.6: dedupStatistics.xml: optimizedFileCount property.

4.6 Next run of the 'Throughput Optimization' job

Source: C:\Windows\System32\Tasks\Microsoft\Windows\Deduplication\ThroughputOptimization. The scheduled task's CalendarTrigger defines a **weekly** schedule on **Thursday**, with a StartBoundary of 2025-09-20. Since the start boundary falls on a Saturday, the first actual execution is the following Thursday at the configured local time, which is then converted to UTC.

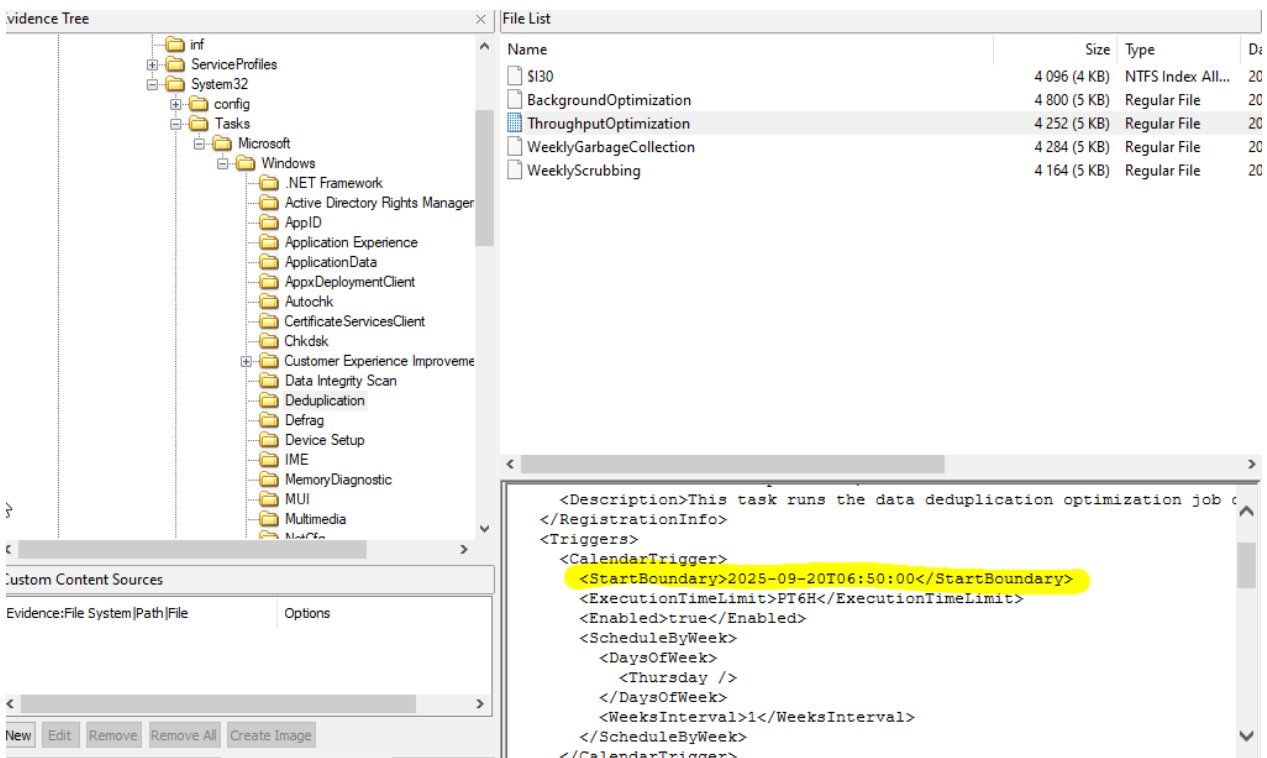


Figure 4.7: ThroughputOptimization task definition: CalendarTrigger, StartBoundary and ScheduleByWeek.

4.7 MFT analysis of the deleted file

Source: D:\\$MFT (512 KB = 512 records). The \$MFT is a file, not a directory, and is therefore located at the root of the D: volume in the FTK Imager file list rather than in the evidence tree.

File List			
Name	Size	Type	Date Accessed
HR	368 (1 KB)	Directory	17/09/2025 03:24:21
IT	56 (1 KB)	Directory	18/09/2025 09:35:33
Legal	56 (1 KB)	Directory	18/09/2025 09:36:59
Procurement	352 (1 KB)	Directory	17/09/2025 03:24:21
Projects	416 (1 KB)	Directory	17/09/2025 03:24:20
System Volume Information	256 (1 KB)	Directory	18/09/2025 14:43:10
\$AttrDef	2 560 (3 KB)	Regular File	16/09/2025 07:55:32
\$BadClus	-	Regular File	16/09/2025 07:55:32
\$Bitmap	16 384 (16 KB)	Regular File	16/09/2025 07:55:32
\$Boot	8 192 (8 KB)	Regular File	16/09/2025 07:55:32
\$I30	4 096 (4 KB)	NTFS Index All...	18/09/2025 13:29:28
\$LogFile	4 341 760 (4 24...	Regular File	16/09/2025 07:55:32
\$MFT	524 288 (512 KB)	Regular File	16/09/2025 07:55:32
\$MFTMirr	4 096 (4 KB)	Regular File	16/09/2025 07:55:32
\$Secure	56 (1 KB)	Regular File	16/09/2025 07:55:32
\$TXF_DATA	56 (1 KB)	NTFS Logged ...	18/09/2025 13:29:28
\$WinCsr	131 072 (128 KB)	Regular File	16/09/2025 07:55:32

00000	46 49 4C 45 30 00 03 00-A6 D8 51 00 00 00 00 00	FILE0---	02
00010	01 00 01 00 38 00 01 00-A0 01 00 00 00 04 00 00	---	8
00020	00 00 00 00 00 00 00 00-07 00 00 00 00 00 00 00	---	7
00030	04 00 FF FF 00 00 00 00-10 00 00 00 60 00 00 00	---	10
00040	00 00 18 00 00 00 00 00-48 00 00 00 18 00 00 00	---	48
00050	36 90 3A 47 DF 26 DC 01-36 90 3A 47 DF 26 DC 01	6-:G&#U-6-:G&#U-	
00060	36 90 3A 47 DF 26 DC 01-36 90 3A 47 DF 26 DC 01	6-:G&#U-6-:G&#U-	
00070	06 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	---	
00080	00 00 00 00 00 01 00 00-00 00 00 00 00 00 00 00	---	0
00090	00 00 00 00 00 00 00 00-30 00 00 00 68 00 00 00	---	30
000a0	00 00 18 00 00 00 03 00-4A 00 00 00 18 00 01 00	---	4A
000b0	05 00 00 00 00 05 00-36 90 3A 47 DF 26 DC 01	6-:G&#U-6-:G&#U-	
000c0	36 90 3A 47 DF 26 DC 01-36 90 3A 47 DF 26 DC 01	6-:G&#U-6-:G&#U-	
000d0	36 90 3A 47 DF 26 DC 01-00 40 00 00 00 00 00 00	6-:G&#U-0	
000e0	00 40 00 00 00 00 00 00-06 00 00 00 00 00 00 00	---	06

Figure 4.8: \$MFT (524,288 bytes) at the root of D:; the 'FILE' record magic is visible in the hex viewer.

Browsing the Finance directory reveals the two annual reports. The **Internal** version carries the deleted-file marker (red cross) and is stored as a Deduplication **Reparse Point**, whereas the **Audited** version is still present.

File List			
Name	Size	Type	Date Accessed
Audited Consolidated Financial Statements 2024.docx	78 323 (77 KB)	Reparse Point	16/09/2025 07:55:32
Audited Consolidated Financial Statements 2024.docx.FileSlack	52 749 (52 KB)	File Slack	16/09/2025 07:55:32
Audited Consolidated Financial Statements 2024.pdf	346 625 (339 KB)	Reparse Point	16/09/2025 07:55:32
Audited Consolidated Financial Statements 2024.pdf.FileSlack	46 591 (46 KB)	File Slack	16/09/2025 07:55:32
Internal Consolidated Financial Statements 2024.docx	90 593 (89 KB)	Reparse Point	16/09/2025 07:55:32
Internal Consolidated Financial Statements 2024.docx.FileSlack	40 479 (40 KB)	File Slack	16/09/2025 07:55:32

00000	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	---	
00010	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	---	
00020	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	---	
00030	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	---	

Figure 4.9: D:\Finance\Reports\Annual\2024: the internal report is marked deleted (reparse point).

The corresponding MFT record has its in-use flag cleared (inUse = false). Its \$STANDARD_INFORMATION flags combine several FILE_ATTRIBUTE bits consistent with a deduplicated file, and its \$REPARSE_POINT attribute (type 0xC0) carries the Dedup reparse tag 0x80000013.

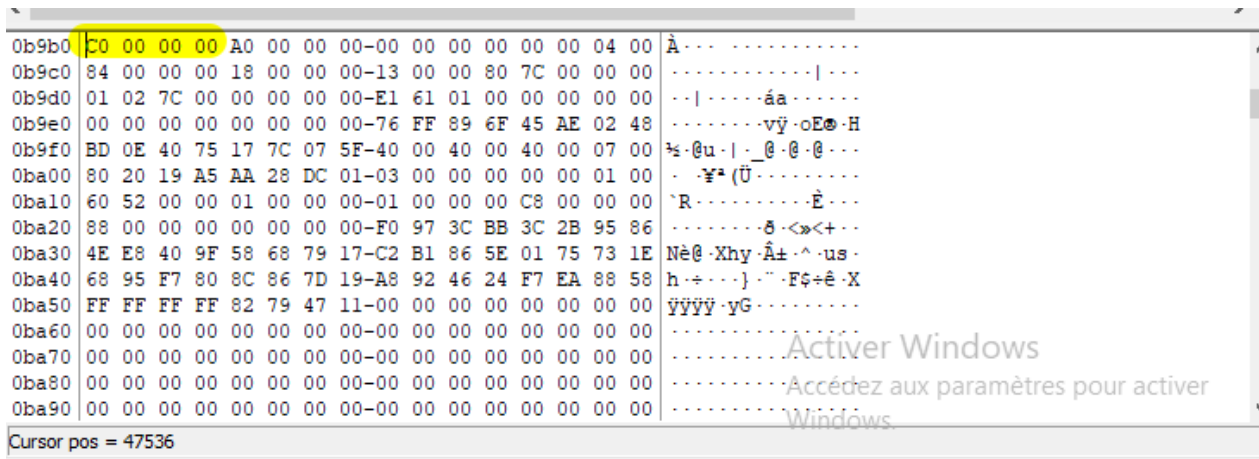


Figure 4.10: \$REPARSE_POINT attribute (0xC0) located within \$MFT; reparse tag 0x80000013 highlighted.

Decoding of the reparse buffer (124 bytes). Offset +0x08: logical size of the file. Offset +0x18: ChunkStore GUID {6F89FF76-AE45-4802-BD0E-4075177C075F}. Offset +0x58: 32-byte stream-map hash, followed by a 4-byte checksum. The file exported via its path is entirely zero-filled (a sparse stub); the real content lives in the ChunkStore and must be reconstructed.

4.8 Reconstruction from the Dedup ChunkStore

Sources: the Deduplication containers under D:\System Volume Information\Dedup\ChunkStore\{6F89FF76-...}.ddp\; the **Stream** container (stream maps) and the **Data** container (chunk data). Both .ccc files begin with the Cthr magic and contain Ckhr (chunk header) and Smap (stream map) records.

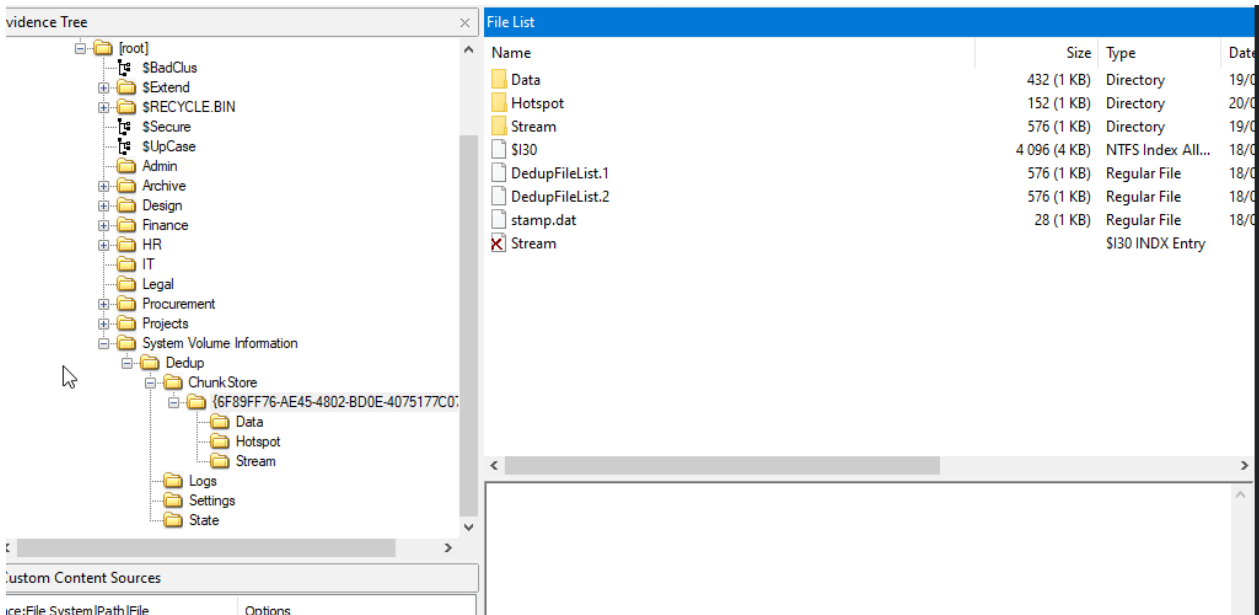


Figure 4.11: ChunkStore '{6F89FF76-...}.ddp' with its Data and Stream sub-directories.

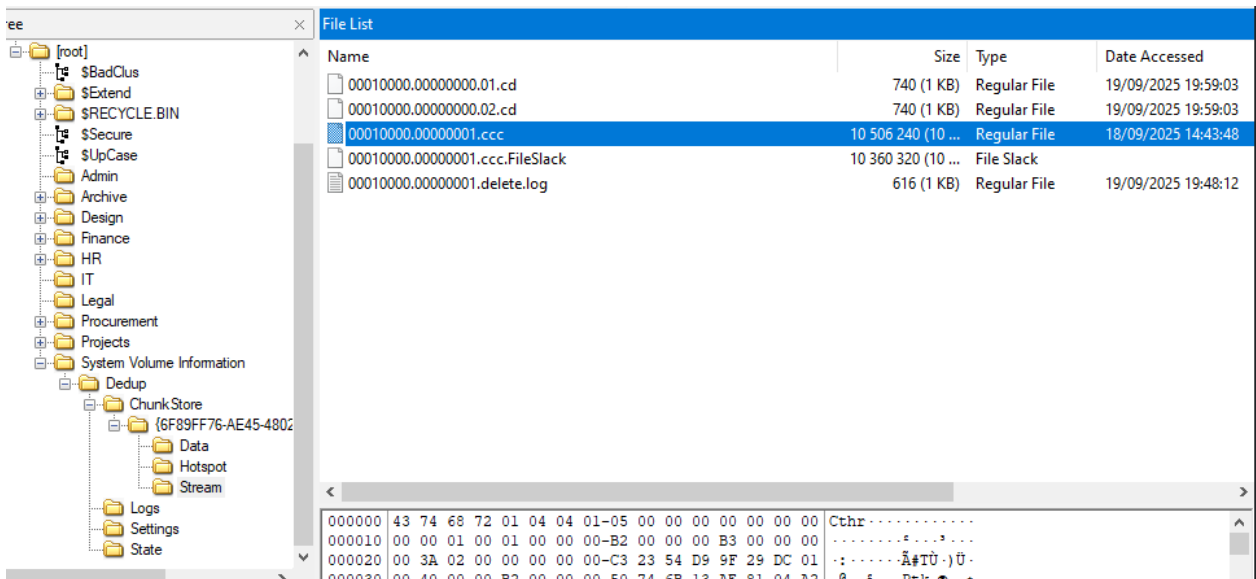


Figure 4.12: Stream container 00010000.00000001.ccc; the 'Cthr' magic is visible in the hex viewer.

Step 1: locate the stream map. The 32-byte reparse hash is searched in the Stream container; it resolves to an Smap record listing, in order, the hash and size of each chunk composing the file. The sum of the chunk sizes must equal the logical size recorded in the reparse buffer.

```
Smap (Internal report) -> N chunk entries
chunk i : hash <32 bytes>   size
...
SUM(size) == logical size (reparse +0x08)
```

Step 2: extract and concatenate. Each chunk is retrieved by hash from the Data container (chunk header Ckhr, data at offset +0x58) and concatenated in order. The first chunk begins with 50 4B 03 04 ('PK', the ZIP/DOCX magic). The chunks are stored uncompressed, consistent with .docx being in the noCompressionFileExtensions list. The reassembled file is then hashed (SHA-256) to fix its identity as evidence.

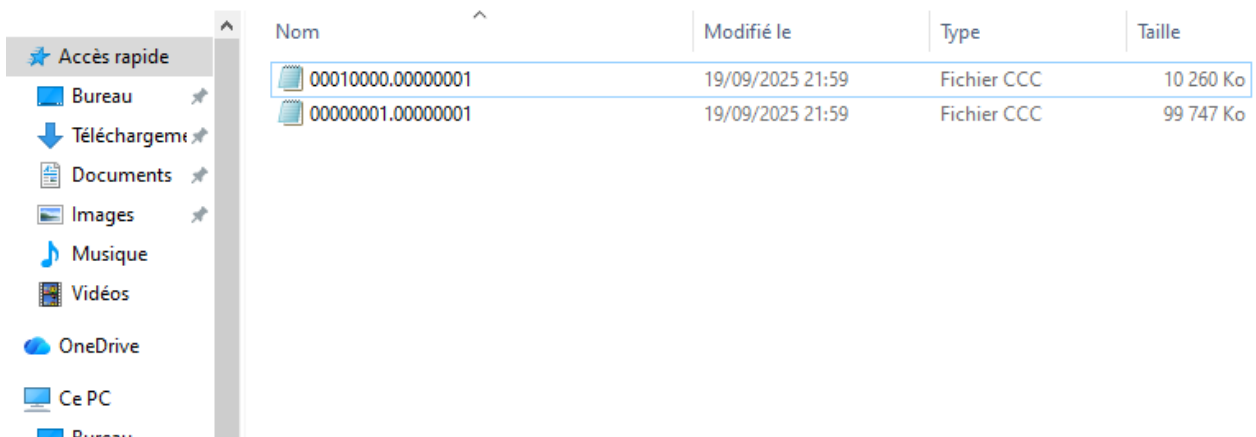


Figure 4.13: Export of the ChunkStore containers from FTK Imager for reconstruction.

4.9 Automated recovery tool (recover.py)

A custom Python tool was developed to automate the reconstruction. It parses the \$MFT to find the deleted record carrying a Dedup reparsing point (tag 0x80000013), extracts the 32-byte stream-map hash directly from the reparsing buffer (no manual input), reads the stream map from the Stream container, then extracts and concatenates the chunks from the Data container, finally computing the SHA-256 of the result.

```

PS C:\Users\vboxuser\Desktop\export> python .\recover.py ".\$MFT" ".\00000001.00000001.ccc" ".\00010000.00000001.ccc" recovered_Internal.docx
46
[*] $MFT : .\$MFT (524288 octets)
[*] Conteneur Data : .\00000001.00000001.ccc (102140416 octets)
[*] Conteneur Stream : .\00010000.00000001.ccc (10506240 octets)

[*] 165 fichier(s) dedupliques trouves dans le $MFT :
    enreg #44 [present ] Audited Consolidated Financial Statements 2024.docx
    enreg #45 [present ] Audited Consolidated Financial Statements 2024.pdf
    enreg #46 [SUPPRIME] Internal Consolidated Financial Statements 2024.docx

```

Figure 4.14: recover.py invocation: the \$MFT scan lists the deduplicated accounting files and flags the Internal report as SUPPRIME (deleted).

```

[*] Cible retenue : enreg #46 [supprime] 'Internal Consolidated Financial Statements 2024.docx'
[*] Hash stream map : f0973cbb3c2b95864ee8409f58687917c2b1865e0175731e6895f7808c867d19
[*] 1376 chunk(s) indexes dans le conteneur Data
[*] Stream map : 2 chunk(s)
    chunk 1 : hash=68e568e2f0f41373... taille=61095
    chunk 2 : hash=5ffbe14ef8342bb7... taille=29498
+ chunk 1 : id=7 taille=61095 debut=504b0304
+ chunk 2 : id=8 taille=29498 debut=8a56e190

[+] Recovered size : 90593
[+] SHA-256 : e51e773e5404f29cc2816cff3fbdcc8b2c28f9e8040a6e7fe926b21336b66513
[+] Ecrit : recovered_Internal.docx
PS C:\Users\vboxuser\Desktop\export> ls

Répertoire : C:\Users\vboxuser\Desktop\export

Mode                LastWriteTime         Length Name
----                -
-a----          19/09/2025    21:59      102140416 00000001.00000001.ccc
-a----          19/09/2025    21:59      10506240 00010000.00000001.ccc
-a----          05/07/2026     01:46         1530 diag_reparse.py
-a----          05/07/2026     01:51        11184 recover.py
-a----          05/07/2026     01:52        90593 recovered_Internal.docx

PS C:\Users\vboxuser\Desktop\export>

```

Figure 4.15: recover.py output: target record, stream map, recovered size and SHA-256 of the reconstructed document.

Fraud indicator. The \$MFT scan reveals that **both** accounting documents are deduplicated: the 'Audited' versions (.docx and .pdf, still present) and the 'Internal' version (**deleted**). The organisation kept the public version and deleted the truthful internal one, which is direct evidence of intent to conceal.

4.10 Authorship of the internal report

Source: docProps/core.xml inside the reconstructed .docx (a .docx is a ZIP archive). The archive was extracted with PowerShell and the core properties (dc:creator, cp:lastModifiedBy, creation and modification dates) were inspected to attribute the document.

```

PS C:\Users\vboxuser\Desktop\export> ls

Répertoire : C:\Users\vboxuser\Desktop\export

Mode                LastWriteTime         Length Name
----                -
-a----          19/09/2025    21:59      102140416 00000001.00000001.ccc
-a----          19/09/2025    21:59      10506240 00010000.00000001.ccc
-a----           05/07/2026    01:46         1530 diag_reparse.py
-a----           05/07/2026    01:51         11184 recover.py
-a----           05/07/2026    01:52         90593 recovered_Internal.docx
-a----           05/07/2026    01:52         90593 recovered_Internal.zip

PS C:\Users\vboxuser\Desktop\export> Expand-Archive .\recovered_Internal.zip -DestinationPath .\ex -Force
PS C:\Users\vboxuser\Desktop\export> Get-Content .\ex\docProps\core.xml
<?xml version='1.0' encoding='UTF-8' standalone='yes'?>
<cp:coreProperties xmlns:cp="http://schemas.openxmlformats.org/package/2006/metadata/core-properties" xmlns:dc="http://purl.org/dc/elements/1.1/"
xmlns:dcterms="http://purl.org/dc/terms/" xmlns:dcmttype="http://purl.org/dc/dcmitype/" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
"><creator>Elaine Chua - Senior Accountant</dc:creator><cp:lastModifiedBy>Elaine Chua - Senior Accountant</cp:lastModifiedBy><cp:revision>1</cp:revision><dcterms:created
xsi:type="dcterms:W3CDTF">2025-09-16T15:59:00Z</dcterms:created><dcterms:modified
xsi:type="dcterms:W3CDTF">2025-09-16T16:02:00Z</dcterms:modified></cp:coreProperties>
PS C:\Users\vboxuser\Desktop\export>

```

Figure 4.16: core.xml of the recovered document: creator and last-modified-by properties.

4.11 Overstatement of the annual profit

The income statements of the two reconstructed versions were compared side by side. The audited (public) version reports a profit for the year, whereas the internal (real) version reports a loss. Revenue, operating profit and profit before taxation are all inflated in the public version; the overstatement of the annual profit is the difference between the published figure and the real one.

WordPad ne prend pas en charge toutes les fonctions de ce format de document. Certains contenus peuvent être manquants ou affichés incorrectement.			WordPad ne prend pas en charge toutes les fonctions de ce format de document. Certains contenus peuvent être manquants ou affichés incorrectement.		
	Note	2024 \$'000		Note	2024 \$'000
Revenue	22	10,082,467	Revenue	22	6,083,427
Materials and subcontract costs		(6,273,001)	Materials and subcontract costs		(7,254,193)
Staff costs	23	(1,432,889)	Staff costs	23	(1,598,204)
Depreciation and amortisation		(208,571)	Depreciation and amortisation		(213,452)
Other operating expenses		(270,699)	Other operating expenses		(287,511)
Operating profit	24	1,897,307	Operating profit	24	-1,269,933
Investment income	25	24,589	Investment income	25	24,589
Interest income	25	113,982	Interest income	25	113,982
Interest expenses	25	(98,230)	Interest expenses	25	(98,230)
Share of results of associated companies	8	239,772	Share of results of associated companies	8	239,772
Profit before tax and exceptional items		2,177,420	Profit before tax and exceptional items		2,177,420
Exceptional items	26	1,135,282	Exceptional items	26	1,135,282
Profit before taxation		3,312,702	Profit before taxation		-1,432,882
Taxation	27	(640,506)	Taxation	27	(214,673)
Profit for the year		2,672,196	Profit for the year		-1,647,555
Attributable to:			Attributable to:		
Shareholders of the Company		1,840,512	Shareholders of the Company		1,840,512
Non-controlling interests		831,684	Non-controlling interests		831,684
		2,672,196			2,672,196

Figure 4.17: Side-by-side income statements. Left: audited/public version. Right: internal/real version.

$$\text{Overstatement} = \text{Profit_for_the_year(audited)} - \text{Profit_for_the_year(internal)} \quad (\$'000)$$

5. Conclusion & Actionable Intelligence

The organisation maintained two sets of 2024 accounts: a public, audited version showing a profit and an internal version revealing a real loss. The internal document was deleted to conceal the losses. Because Windows Data Deduplication remained active on the D: volume, the deleted file's data survived in the ChunkStore and was fully reconstructed and validated by SHA-256. Comparison of the two versions demonstrates a material overstatement of both revenue and annual profit.

5.1 Indicators

- **Primary artefact:** reconstructed internal 2024 Consolidated Financial Statements (.docx), to be preserved as evidence together with its SHA-256.
- **Deletion of material evidence:** MFT record of the internal report marked deleted while the falsified public version was retained, indicative of intent to conceal.
- **Dedup residue:** reparse point (tag 0x80000013) and ChunkStore stream map still referencing the deleted content.

5.2 Recommendations

- Preserve the SYSTEM hive, the Dedup artefacts (Settings, State, Logs) and the full ChunkStore; avoid any garbage-collection job on D: that could purge orphaned chunks.
- Extend the review to prior fiscal years and to the HR directory.
- Correlate with account access logs and file-system journals (\$LogFile, \$UsnJrnl) to attribute the deletion.
- Hand over the reconstructed document and hash values to legal/compliance under documented chain of custody.